Reg No: **Name:**

Government Engineering College Palakkad

Department of Information Technology

Assignment 1

Subject Name: Discrete Mathematical Structures

1. Find the dual of the compound prepositions
   a) $P \wedge Q \vee \neg R$ b) $(P \vee F) \wedge (Q \vee T)$
2. Show that $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent.
3. Use De Morgan's laws to find the negation of each of the following statements.
   a) Ananya is rich and happy.
   b) Bijoy will bicycle or run tomorrow.
   c) Remya is smart and hardworking.
   d) Vignesh walks or takes the bus to class.
4. Let C(x) be the statement "x has a cat". Let D(x) be the statement "x has a dog", and F(x) be the statement "X has a ferret". Express each of the following statements in terms of C(x), D(x), and F(x), quantifiers, and logical connectives. Let the domain consist of all students in your class.
   a) A student in your class has a cat, a dog, and a ferret.
   b) All students in your class have a cat, a dog, and a ferret.
   c) Some students in your class have a cat and a ferret, but not a dog.
   d) No student in your class has a cat, a dog, or a ferret.
   e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has one of these animals as a pet.
5. Rewrite the statements so that negations appear only within predicates (i.e so that no negation is outside a quantifier or an expression involving logical connectives)
   a) $\neg\exists x \exists y P(x,y)$
   b) $\neg\exists y (Q(y) \wedge \forall x \neg R(x,y))$
   c) $\neg\exists y(\exists x R(x,y) \vee \forall S(x,y))$
6. Explain why (A X B) X (C X D) and A X (B X C) X D are not the same.
7. Determine whether each of the functions is a bijection from R to R.
   a) $f(x)=2x+1$
   b) $f(x)= (x^2 + 1)/ (x^2 +2)$

8. Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x+2$, are functions from R to R.

9. Let R be the relation represented by the matrix

$$M_R = \begin{matrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{matrix}$$ Find the matrix representing R^{-1}, R^2, and $\bar{R}$.

10. Answer these questions for the poset ({2,4,6,9,12,18,27,36,48,60,72}, I).

a) Find maximal elements.

b) Find minimal elements.

c)Is there a greatest element?

b) Is there a least element?

d) find all upper bounds of {2,9}.

e) Find LUB and GLB of {2,9} and {60,72}

11. Check validity of the statement

$p \rightarrow q$

$q \rightarrow (r \wedge s)$

$\neg r \vee (\neg t \vee u)$

$p \wedge t$

$\therefore u$

12. Let p, q, and r are the following statements:
p: Ajai studies. q: He plays volleyball. r: He passes programming in C
let P1, P2, and P3 are the premises.
P1: If Ajai studies, then he will pass programming in C.
P2: If he does not play volleyball, then he will study.
P3: He failed in programming in C.
Determine whether the argument $(P1 \wedge P2 \wedge P3) \rightarrow q$ is valid